



OPEN Probing the diversity in dairy cattle reticulorumen temperature for adaptation selection

A. K. Shirley^{1✉}, P. C. Thomson², A. Chlingaryan¹ & C. E. F. Clark³

There remains a paucity of literature investigating the direct link between variability in dairy cattle core body temperature response to thermal indices, despite advances in precision livestock monitoring through sensor technology. Here we decomposed thermal indices and reticulorumen temperature data for 1,429 dairy cattle across three dairy farms in Victoria, Australia to investigate the variability in individual response to climate at different temperature humidity index (THI) thresholds. We revealed an overall positive association between THI and reticulorumen temperature deviation, with significant variability in sensitivity to thermal change among individuals. Increases in reticulorumen temperature were evident at a THI_{max} of 65, indicating a review of industry-established THI thresholds are required. Baseline THI levels were shown to be a major influencing factor for animal response. From these insights, our work is now focused on the development of a core body temperature derived phenotype for the selection of heat-tolerant animals.

Heat stress (HS) poses a significant challenge for dairy cattle welfare, and with increasing global temperatures, the prevalence and severity of HS among dairy herds is expected to increase. Dairy cattle, particularly in pasture-based systems, face direct exposure to solar radiation¹ and fluctuating environmental conditions², making them vulnerable to climate factors. While environmental indices such as the temperature-humidity index (THI) are widely used to predict HS, they represent an external measure of heat load. More importantly, we must account for individual variability in response among animals. This inter-animal variability is increasingly recognised as a key influencing factor in the effectiveness of heat mitigation strategies^{3,4}.

Monitoring reticulorumen temperature offers a promising method to supervise internal heat load in dairy cattle, providing a direct, continuous measure of the body's response to environmental stress^{5,6}. By capturing real-time data on an individual level, reticulorumen temperature can reveal cattle response to HS under varying THI thresholds, providing insights into individual heat tolerance. Correlations between ruminal and reticular temperatures and the ambient thermal environment have been demonstrated^{6–9}, validating the use of reticulorumen sensors in dairy cattle. These sensors offer a non-invasive, scalable alternative to vaginal and rectal methods, which – despite their widespread use in characterising HS responses – are limited by infection risk, gender constraints, and reduced suitability for commercial application^{10–12}. Despite the validation of reticulorumen sensors, little is known about the extent to which THI correlates with reticulorumen temperature across individual animals in commercial settings. A THI threshold of 72 is commonly cited as a critical value, above which production decline will occur^{13–16}, yet such generalities do not capture the nuanced response of individual cattle.

Understanding the relationship between THI and reticulorumen temperature is essential for improving HS management strategies. Identification of individual variability in response to heat might allow for more tailored interventions in the short term, and dedicated breeding strategies in the long term, ensuring that heat mitigation efforts are both effective and efficient. While there is extensive research on the general effects of HS on dairy cattle^{17,18}, understanding the interplay of individual animal variability in reticulorumen temperature and THI deviation is less explored, particularly in pasture-based systems where cattle experience a high degree of environmental variation.

In this work, we continuously recorded reticulorumen temperature data (smaXtec Animal Care GmbH) from 1,429 dairy cattle across three farms in Victoria, Australia. To characterise reticulorumen temperature, fluctuations attributable to drinking events were isolated and removed to focus the analysis on heat-load related

¹Livestock Production and Welfare Group, School of Life and Environmental Sciences, University of Sydney, Camden, NSW 2570, Australia. ²Sydney School of Veterinary Science, University of Sydney, Camden, NSW 2570, Australia. ³Gulbali Institute, Charles Sturt University, North-Wagga Wagga, NSW 2650, Australia. ✉email: alice.shirley@sydney.edu.au

patterns rather than transient effects of water intake. Climatic data for the same period were interpolated from surrounding weather stations to calculate THI. The association of reticulorumen temperature with simultaneously collected THI was explored with raw data and after removing diurnal patterns and long-term trends from the time-series to investigate the variability in HS response at different THI thresholds. The objective of this study was to investigate individual variation in reticulorumen temperature response to thermal indices in pasture-based dairy cattle, with the hypothesis that sensitivity to HS varies significantly among animals and is detectable at lower THI thresholds than currently recognised. This work has the potential to contribute to the development of proactive and individualised strategies to enhance dairy cattle welfare and productivity in increasingly challenging conditions.

Results

Overall positive association between THI and reticulorumen temperature deviation, with notable variability in individual sensitivity

Following de-trending of both THI_{max} and reticulorumen temperature time series, the association between THI_{max} (deviation; irrespective of baseline) and reticulorumen temperature deviations ($^{\circ}C$) was assessed (Fig. 1). An overall positive association was observed at both the farm and individual levels. Farm B depicted greater sensitivity as indicated by the overall steeper slope compared to Farms A and C, acknowledging its geographical location further north (southern hemisphere). At the individual level, the visualisation highlights significant variability in sensitivity to thermal change among individuals (Supplementary Table 2), with individual variability less pronounced for animals from Farms A and C. Cross-correlation analyses (Supplementary Fig. S1) confirm correlation was highest on Day 0 although significant positive associations were detected for the next two days following a change in THI_{max} . Given the peak correlation on Day 0, lag terms were not carried through for model analysis. A Wald test indicated significant differences between farms ($p = 0.0057$), THI_{max} (deviation) ($p < 0.001$), as well as a farm by THI_{max} (deviation) interaction ($p < 0.0001$) (Supplementary Table 1). Differences in individual sensitivity to a change in reticulorumen temperature per unit change in THI were evident, as estimated by the random slopes in the mixed model (Fig. 2). All animals maintained an average reticulorumen temperature change below $0.035^{\circ}C/THI$ unit, ranging from below 0 to 0.030.

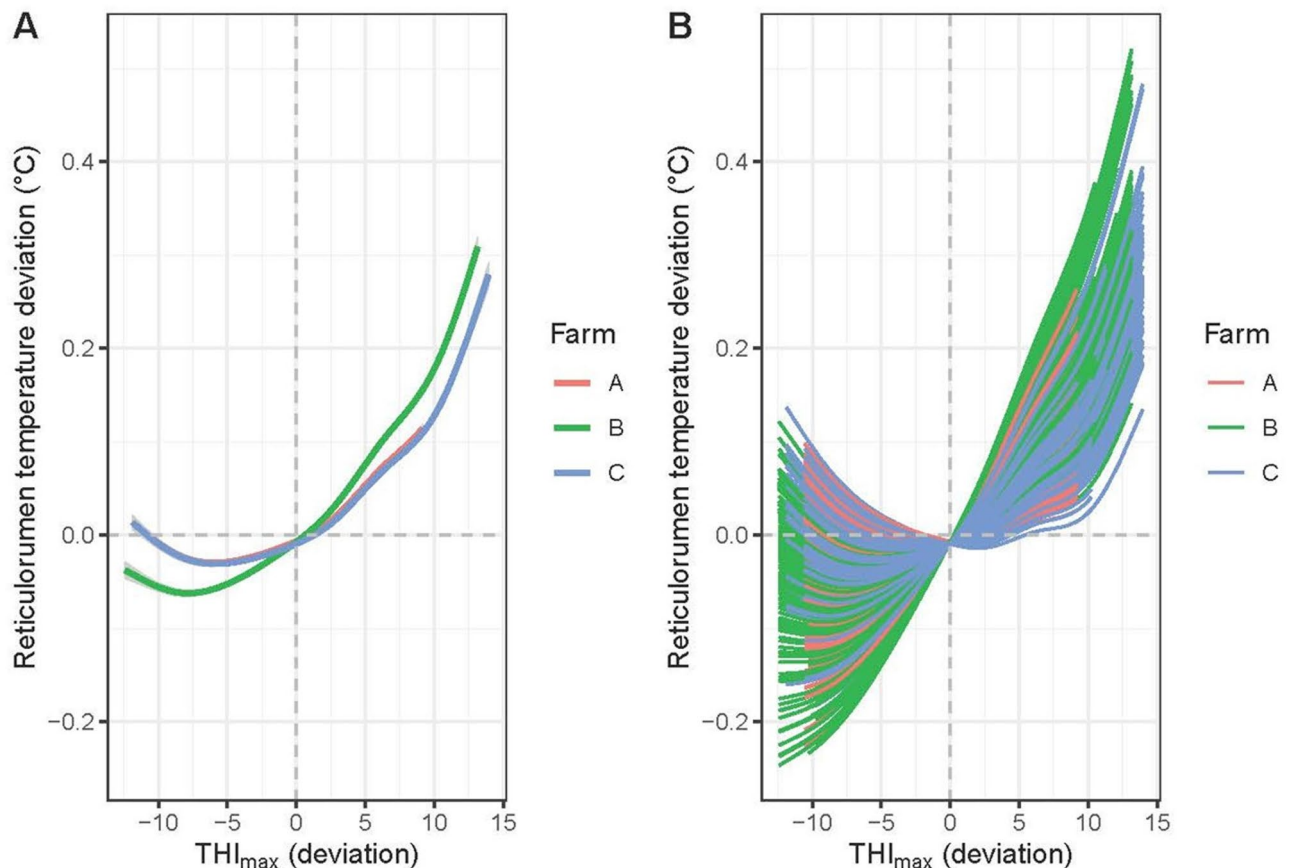


Fig. 1. Association between THI_{max} (deviation) and reticulorumen temperature deviation ($^{\circ}C$). Association depicted at both the (A) farm and (B) individual level. THI_{max} (deviation) is the effect of THI deviation regardless of baseline. Truncation applied to display only THI ranges experienced by the farm and/or individual. The grey band in (A) represents ± 1 standard error.

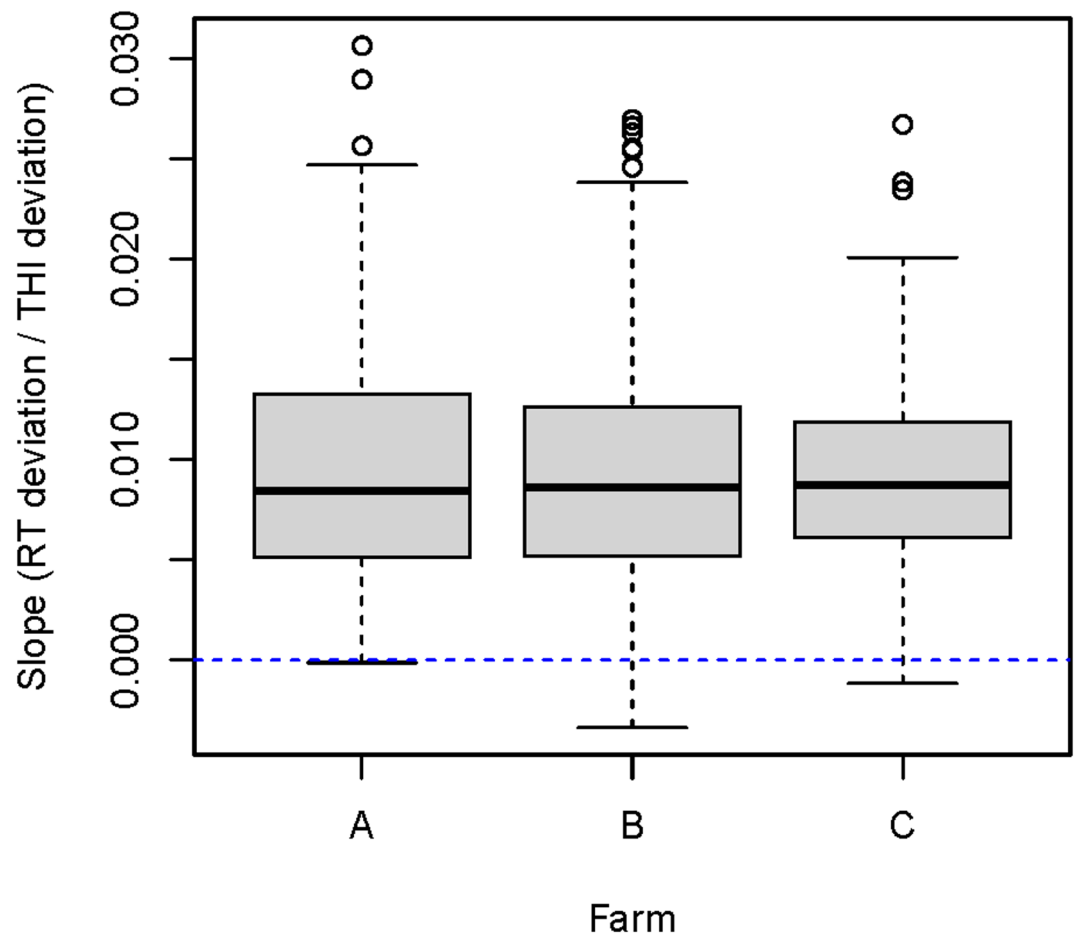


Fig. 2. Sensitivity of reticulorumen temperature (RT) deviation on THI deviation for each animal. Slopes of the individual animal reticulorumen response per change in THI deviation, for each animal.

Baseline THI levels are a major influencing factor for animal response

The association between THI_{max} (smooth; accounting for the initial THI baseline) and reticulorumen temperature deviation ($^{\circ}C$) at both the farm and individual levels was assessed (Fig. 3). The data revealed an increase in reticulorumen temperature at a THI_{max} (smooth) of approximately 65, corresponding to minimum points observed at 64.9, 65.2, and 64.3 for Farms A, B, and C respectively. This was followed by a second, more marked increase at a THI_{max} (smooth) of approximately 75. Based on Wald tests, there were significant differences between farms ($p < 0.0001$), THI_{max} (smooth) ($p < 0.0001$), and farm by THI_{max} (smooth) interaction ($p < 0.0001$) (Supplementary Table 3). The individual animal plot closely mirrors the farm-level plot, indicating minimal detectable variation at the individual level. This lack of variation was confirmed by the estimated variance components in the mixed model (Supplementary Table 4).

Association between THI_{max} (deviation) and THI_{max} (smooth) on reticulorumen temperature deviation ($^{\circ}C$), for each farm was evaluated (Fig. 4). Reticulorumen temperature exhibited a stronger positive response with increasing THI_{max} (smooth). This effect was more pronounced at higher THI_{max} deviation values, indicating a compounding influence of baseline THI on physiological response to variations. At both positive and negative THI_{max} (deviation) extremes, if THI_{max} (smooth) was low, a paradoxical response of reticulorumen temperature was evident. While the overall trend was consistent across farms, there were subtle differences in the magnitude of response. The standard error of prediction has been accounted for in a complementary surface plot (Supplementary Fig. S2).

Discussion

Here we reveal the impact of changing thermal conditions on reticulorumen temperature through time series and mixed model analysis. The decomposition of THI_{max} enabled removal of long-term and seasonal trends, facilitating an assessment of how deviations in THI impact subsequent animal response. With minor deviations in THI_{max} , all cattle were able to maintain similar reticulorumen temperature, indicating a thermoneutral zone. Different overall levels of temperature response were evident at the farm level in response to larger THI deviations, with individuals shown to vary significantly in their sensitivity. Also conducting a time series decomposition, Islam, et al.³ presented similar results in feedlot beef cattle, demonstrating an overall increase in core body temperature with increasing THI, identifying a distinction in response between ‘susceptible’ and ‘tolerant’ cattle

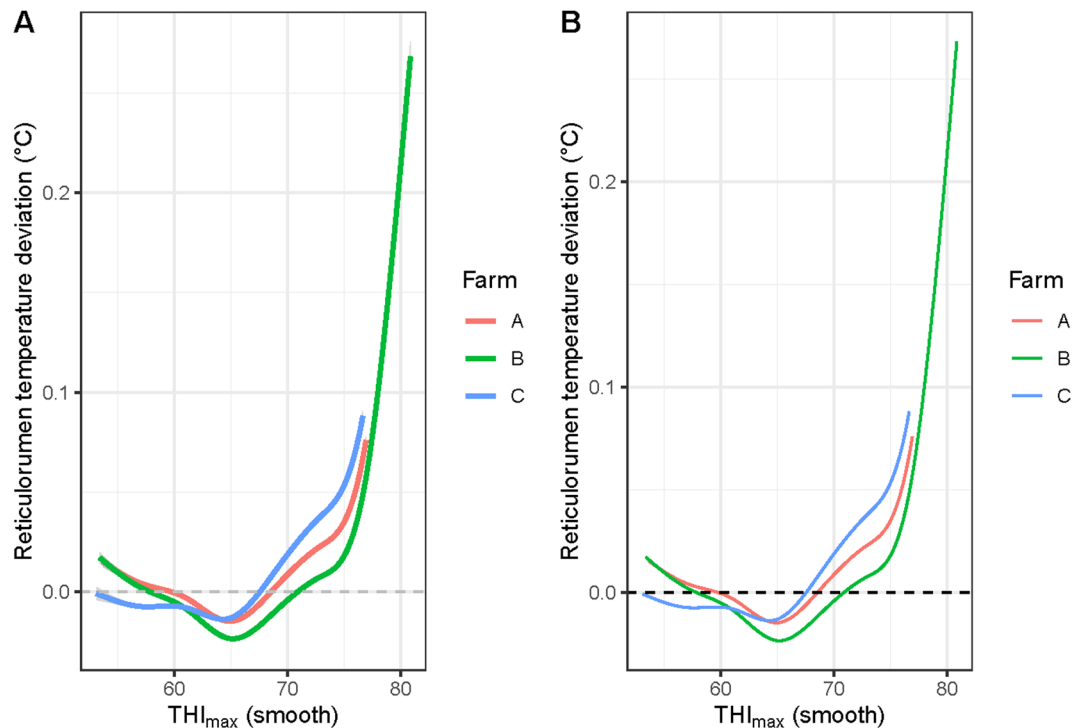


Fig. 3. Association between THI_{max} (smooth) and reticulorumen temperature deviation ($^{\circ}\text{C}$). Association depicted at both the (A) farm and (B) individual level. THI_{max} (smooth) is the effect of THI deviation accounting for baseline value. Truncation applied to display only THI ranges experienced by the farm and/or individual. The grey band in (A) represents ± 1 standard error.

beyond a THI range of 70–74. Here, at both the farm and individual level, animals on the most northern property were most affected by THI_{max} deviations. Dairy cows retain a known sensitivity to high ambient temperature^{19,20}, therefore temperature deviations might have greater impact for animals experiencing warmer conditions. While THI remains a widely used indicator of HS, its omission of key environmental factors, namely solar radiation and wind speed, might limit its accuracy in pasture-based systems; future studies should explore the application of more comprehensive indices, such as the dairy heat load index²¹ or the grazing heat load index²², that are specifically developed for lactating and pasture-based dairy cattle. Similarly, under grazing conditions, diurnal grazing behaviour and heterogeneous access to shade and water can introduce asynchronous patterns between ambient thermal indices and internal temperature dynamics. As detailed behavioural and paddock-level environmental data were not available in this study, such factors could not be explicitly accounted for and might contribute to residual variation in reticulorumen temperature responses. Future studies integrating sensor-based temperature measurements with behavioural and environmental metadata would enable more refined interpretation of thermal phenotypes under grazing conditions.

Exploration of individual animal slopes suggests that the group as a whole demonstrated a high level of thermoregulatory capacity, while also highlighting that some individuals still undergo a greater physiological change to rising THI in comparison to others. Analysis of these slopes represents an initial step towards characterising individual heat-response patterns that might inform the development of a core body temperature derived phenotype for the selection of heat-tolerant animals. However, the utility of such phenotypes for genetic selection will depend on their repeatability and robustness, as short-term physiological variability can introduce noise that reduces selection accuracy. Other work has separated animals based on their level of susceptibility³, and this concept provides a useful framework for future refinement and should be introduced here. Further research should therefore explore the repeatability of these responses, that is, whether the same animals consistently demonstrate similar responses across multiple HS events. Rather than singularly exploring how an animal responds to THI deviations, we must consider how consistently that individual can maintain physiological stability across different conditions.

Based on the analysis of THI_{max} (smooth), baseline THI levels were shown to be a major influencing factor for animal response. Variation at the baseline remained significant at the population level. A distinct increase in reticulorumen temperature deviation at a THI_{max} (smooth) of approximately 65 was evident, followed by a second, more marked increase at THI 75. Several studies have confirmed THI predictions are underestimating the physiological impact of HS^{23,24}. This study, therefore, aligns with recommendations by others that heat mitigation should be enacted earlier – at minimum upon exposure of THI 65 and above. A review of industry-established THI thresholds is required to better reflect responses of the high-producing dairy cow seen today. Alternatively, acknowledgement that THI thresholds might be most effectively determined for an individual herd (ideally for individuals within this herd), allowing for a more tailored approach that will account for animal

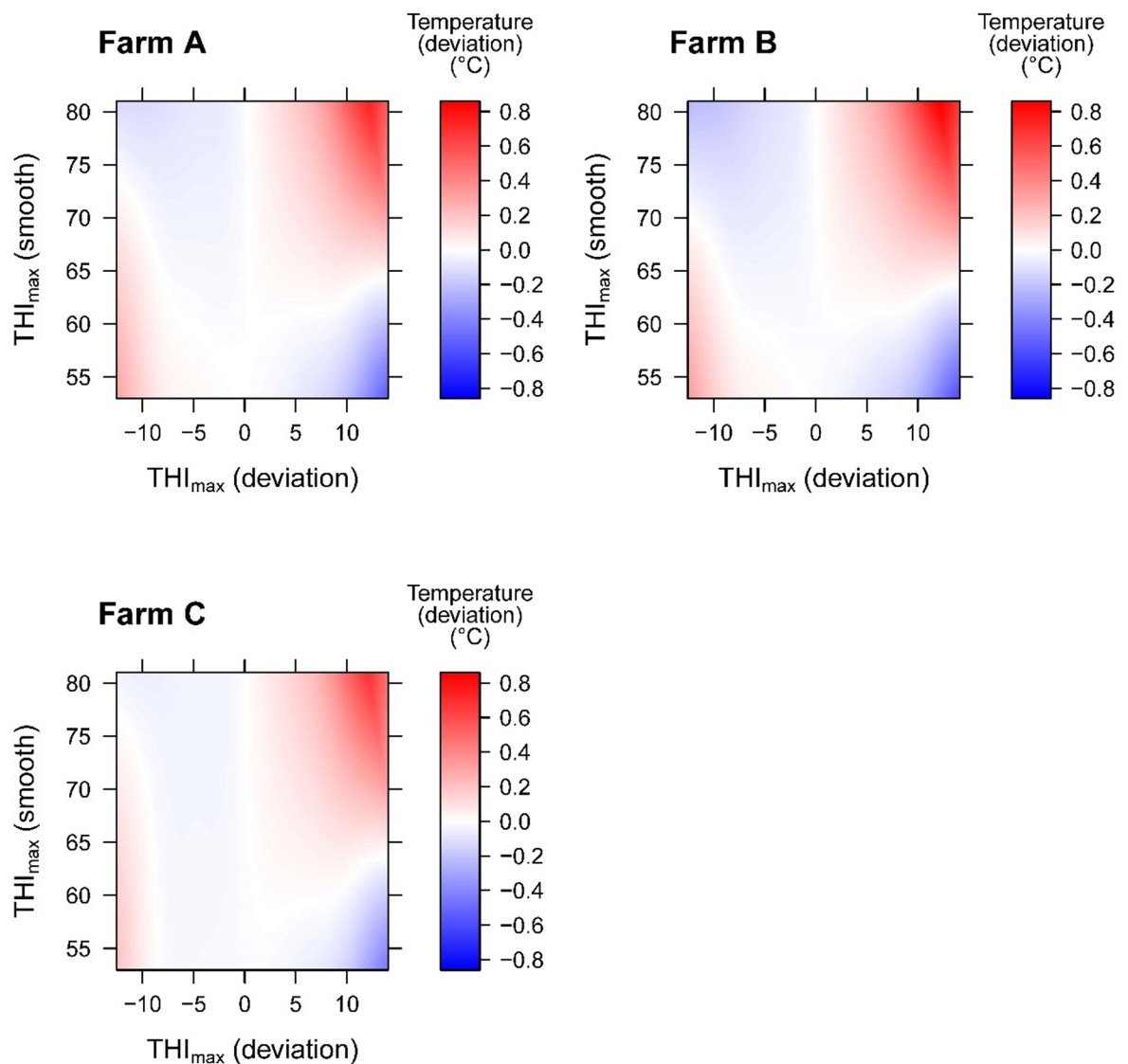


Fig. 4. Reticulorumen temperature response to THI deviations. Surface plot illustrating the relationship between reticulorumen temperature and two measures of THI deviation (THI_{max} (deviation) and THI_{max} (smooth)). An individual surface plot is displayed per farm alongside a colour legend.

factors. Interestingly, minimal detectable variation was evident at the individual animal level for baseline THI_{max} (smooth). We propose three hypotheses in response:

- 1) Environmental conditions exert such a dominant influence at certain thresholds that intrinsic differences are overridden. Yet, this does not explain the maintained lack of variability seen at low thresholds.
- 2) As all individuals (at the farm level) are experiencing the same macro-environment, long-term times measures will appear to have a near identical impact. Analysis at a more regular time interval (i.e. hourly) might demonstrate differing results.
- 3) A standardised response is seen at farm level due to environmental acclimation among individuals.

Individual variation in heat tolerance and susceptibility might impact apparent associations between core body temperature and thermal indices over time^{3,25,26}. In particular, productive stage is known to influence thermal sensitivity, with lactating cows generally exhibiting lower HS thresholds than dry cows. Animals in the present study were modelled as a heterogeneous commercial population due to the retrospective and de-identified nature of the dataset, which precluded consistent inclusion of physiological status. Further research should include greater consideration for individual cow-related factors such as breed, age, and parity that might better reveal animal sensitivity and account for the remaining variability in the presented models.

Mixed models using thin-plate splines have been employed to produce surface plots that show the overall mean response for each farm, though theoretically, each individual has their own response pattern. A paradoxical

	Farm A	Farm B	Farm C
Number of animals	659	426	344
Recording start date	8 June 2021	28 March 2019	7 March 2019
Data download cut-off	2 February 2023	2 February 2023	2 February 2023

Table 1. Individual farm statistics.

response of reticulorumen temperature was evident if THI_{max} (smooth) was low, at both positive and negative THI_{max} (deviation) extremes. We suggest this might be indicative of:

- 1) Overcompensation of thermoregulation at large THI deviations.
- 2) Increased effectiveness of employed thermoregulation efforts at low THI baselines.
- 3) Carry over effect from temperature extremes (both hot and cold) from preceding days.

Highlighted here is the influence that the starting THI has on the resultant reticulorumen temperature impact. If an animal is positioned at lower starting THI baseline, it has greater capacity for thermoregulation, therefore a lower effect on reticulorumen temperature is evident. Conversely, if an animal is exposed to a higher THI baseline, there is reduced ability to buffer an increase in environmental temperature and a greater effect on reticulorumen temperature is seen. Cowan, et al.²⁷ have identified the requirement for an equivalent HS alert for cattle to the Australian Bureau of Meteorology sheep graziers alert, highlighting the benefits of using multiple thermal stress indices and standardising resultant risk classifications. We emphasise the requirement for all HS alerts to account for the preceding environmental conditions (i.e. starting THI) and not just those predicted, to most accurately forecast the resulting impact on cattle.

Methods

This research was conducted using historical reticulorumen temperature data from 1,429 dairy cows across three farms in Victoria, Australia from June 2021 ongoing, with a data download cut-off for ease of analysis in February 2023 (Table 1). The study region is characterised by temperate climatic conditions, with winter-dominant rainfall and high, consistent flows in later winter-early spring. Animal ethics approval was not required as the data were already being recorded by the farms for their own use, as such all experiments were performed in accordance with relevant guidelines and regulations.

All farms were managed under commercial pasture-based grazing systems typical of the region, with herds comprising animals of varying age, breed (predominantly Holstein–Friesian), and parity. Data were retrospectively sourced from routine commercial farm records, and due to data de-identification, detailed animal demographic information (e.g., parity, lactation stage) and farm management details were not consistently available. Similarly, detailed information on paddock configuration, shade availability, tree cover, irrigation infrastructure, and fine-scale grazing behaviour could not be obtained. Cattle had been previously orally administered with a bolus, which naturally migrated to the reticulorumen. Farm management remained under individual producer discretion. All animals were retained to ensure representation of commercial operating conditions.

Reticulorumen temperature data

Reticulorumen temperature data was obtained from the smaXtec bolus (<https://smaxtec.com>) as raw reticulorumen temperature data (‘temp’) at 10-min intervals through wireless transmission, enabling initial visualisation of raw data (Supplementary Fig. S3–S4). Additional data included activity (‘act’) and rumination (‘rum_index’) which were not utilised for this study. Product specification states temperatures between 20 and 60 °C can be measured with an accuracy of ± 0.01 °C²⁸. A Python script was used to automatically download the raw bolus data through a provided application programming interface. To maintain a more accurate representation of normal reticulorumen temperature trends, drinking events were identified and isolated as per Shirley et al.²⁹ based on characteristic short-duration temperature decreases associated with water intake and removed; no data were excluded based on time of day. Minimal data processing was conducted, limited to the removal of physiologically implausible reticulorumen temperature observations. To align with other data, a daily average reticulorumen temperature for each individual was determined for use in subsequent analysis.

Climate data

Climatic data were collected at three-hourly intervals from the nearest surrounding Bureau of Meteorology (BOM, <http://www.bom.gov.au/>) weather stations (average distance of 36 km), and the average values of these surrounding stations taken as to best predict on-site weather conditions at each property. From these data, the THI was calculated using the equation from Yousef³⁰.

$$THI = T_{air} + 0.36 \times T_{dp} + 41.2$$

where T_{air} is the dry-bulb temperature (°C) and T_{dp} is the dew point temperature (°C). The constant (41.2) adjusts the index to a commonly accepted scale. The daily maximum (THI_{max}) was then determined for use in subsequent analysis.

Statistical analyses

To investigate the association of THI_{max} and reticulorumen temperature, the following analyses were conducted. To avoid spurious associations, regardless of underlying functional association, time series trends were decomposed to remove long-term and seasonal trends (Supplementary Fig. S5). The principal behind this is to assess how deviations in THI_{max} affect subsequent deviations in body temperature. For each farm, THI data were extracted and converted into a time series with a yearly frequency, before applying seasonal-trend decomposition using the 'stl' function in R.

$$THI_{max} = THI_{max_seasonal} + THI_{max_trend} + THI_{max_dev}$$

where THI_{max} is the observed value on the day, $THI_{max_seasonal}$ reflects the yearly seasonal trends, THI_{max_trend} captures the long-term trend, and THI_{max_dev} captures short-term residual deviations. Seasonal and trend values were summed to produce a 'smoothed' series.

$$THI_{max_smooth} = THI_{max_seasonal} + THI_{max_trend}$$

For each farm, a data set of smoothed THI_{max} and THI_{max_dev} deviations was created for subsequent analysis. The role of the 'smoothed' version of THI_{max} is to provide a 'baseline' value, free of daily fluctuations.

To analyse the reticulorumen temperature data from each farm, the order of unique ID levels was randomised and divided into batches ($n = 10$) for ease of analysis, as all the raw data could not be simultaneously analysed for the decomposition process as described below. For each batch, a mixed-effects model was fitted, using the ASReml package in R.

$$Temp = \beta_0 + \left(\beta_1 + \beta_1^{(Farm)} \right) cDate + Farm + s(cDate) + ID + s(cDate, ID) + \epsilon$$

where Temp is the observed body temperature, cDate is the date centred around its mean for computational stability, Farm and their interaction are fixed effects, random effects for animal ID account for individual variability, and a spline for Date, $s(cDate)$, was nested within ID, $s(cDate, ID)$, to capture individual-specific nonlinear temporal trends. From the fitted model, residuals were obtained, as these were the deviations ($Temp_{dev}$) with trend removed, and carried forward for analysis. These deviations were appended to the batch data which were then combined for further analysis. The use of ASReml over the 'stl' function allowed for simultaneous modelling of fixed effects and complex random effects, providing a more accurate representation of individual differences in the data.

Cross-correlation analysis was conducted to evaluate the temporal relationship between THI_{max} deviations and reticulorumen temperature deviations to assess any lagged responses to changes in THI_{max} . Mixed effects models were used to explore the associations in more detail, with model fitting conducted using ASReml-R. Fixed effects for the model exploring the relationship between reticulorumen temperature and THI_{max} were the interaction between Farm and THI_{max_dev} deviation. Random effects were included for ID (nested with THI_{max_dev} deviation) and spline smoothing terms for each individual. The specification of the model is as follows:

$$Temp_{dev} = \beta_0 + \left(\beta_1 + \beta_1^{(Farm)} \right) THI_{max_dev} + Farm + s(THI_{max_dev}) + ID + s(cDate, ID) + \epsilon$$

where $Temp_{dev}$ is the reticulorumen temperature deviation, after removing the trend, Farm and THI_{max_dev} are the fixed effects, together with their interaction, with random effects for ID to account for individual variability, and a spline for THI_{max_dev} nested within ID to capture individual-specific nonlinear responses to the THI deviations. The model for THI_{max_smooth} was fitted in the same manner, with replacement of THI_{max_dev} with THI_{max_smooth} .

The Wald test was used to assess the significance of the fixed effects, and the variability between individuals was evaluated using estimated variance components. For both models, predicted values were computed from the fitted models, with predictions made over a grid of THI_{max_dev} or THI_{max_smooth} values for each farm as well as for each individual animal within its farm.

A similar model was developed to investigate the effect of simultaneous effects of deviations and smooth THI_{max} , and their interaction, on reticulorumen temperature deviation. This allowed an assessment of how the effect of THI_{max} deviations changes according to its baseline THI_{max} , as given by THI_{max_smooth} . The model was specified as:

$$Temp_{dev} = \beta_0 + \left(\beta_1 + \beta_1^{(Farm)} \right) THI_{max_dev} + \left(\beta_2 + \beta_2^{(Farm)} \right) THI_{max_smooth} + Farm + s(THI_{max_dev}, THI_{max_smooth}) + s(THI_{max_dev}, THI_{max_smooth}, ID) + \epsilon$$

where the term $s(THI_{max_dev}, THI_{max_smooth})$ indicates a 'thin plate' spline. As above, Wald tests and variance components were evaluated, followed by predicted values being computed from the fitted model and merged with farm-specific THI_{max} limits.

Visualisations of model predictions and the thin-plate spline were obtained using the ggplot2 package and the levelplot function in the lattice package, respectively.

Data availability

To retain farm anonymity, producers were assured raw data would remain confidential and would not be shared. The corresponding author may be contacted for any queries relating to the data.

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Author contributions

A.K. Shirley: Conceptualization, Data curation, Formal analysis, Methodology, Visualization, Writing – original draft. P.C. Thomson: Conceptualization, Data curation, Formal analysis, Methodology, Software, Supervision, Visualization, Writing – review & editing. A. Chlingaryan: Conceptualization, Data curation, Formal analysis, Methodology, Supervision, Writing – review & editing. C.E.F. Clark: Conceptualization, Funding acquisition,

Methodology, Project administration, Supervision, Writing – review & editing.

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Declarations

Competing interests

The authors declare no competing interests.

Additional information

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Correspondence and requests for materials should be addressed to A.K.S.

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